is that the future of the facility and of any new experiments depends as much on international politics as on any scientific considerations. During my visit, the hot topic at the cafeteria was not some exciting new experimental result, but a series of back-and-forth newspaper editorials debating the merits of CERN's proposal to build the so-called Future Circular Collider (FCC), a particle collider so big that the 27-kilometer LHC would become merely a pre-accelerator to bring the protons up to a speed where they could begin to circulate in the FCC ring. The FCC could reach energies of 100 TeV, which is about an order of magnitude higher than what's currently possible at the LHC.

As Freya Blekman pointed out to me during my visit, these experiments take decades to set up, and the data from current experiments can take equally long to analyze, so discussions of the next experimental direction have to happen now. The kind of data we are already getting with the LHC and its upcoming upgrades will take us ten or even fifteen more years to fully analyze. "So this is the time to decide," Blekman says. "What do we want? Do we want an electron-positron collider? Should it be linear? Should it be circular? What are the pros and cons of each? Do we want to directly go to a higher-energy proton-proton machine?"

The arguments for and against future colliders, especially the ambitious FCC, can get rather heated. Even if you set aside the cost (around 10 billion euros at a minimum), debates remain around the promise—or lack thereof—that a bigger collider will find new particles. It may be that the elusive "new physics" we're searching for only shows up at energies so high that even gargantuan machines like the FCC have no hope of ever reaching them. Or it may be that just focusing on increasing the energy puts us on the wrong track entirely, and there's some clue about new physics hiding in another regime that we have yet to explore, perhaps even in data we already have.

Researchers I spoke to at CERN were adamant that increasing energy is essential to move us forward, even if only to better understand the Standard Model. Which does, after all, present us with the specter of vacuum decay. If that Sword of Damocles is going to be hanging over our heads, it would be nice to know exactly what it's doing up there.

André David, an LHC researcher in the Compact Muon Solenoid (CMS) collaboration who hosted my visit to the detector, pointed out that answering this question is a key motivation for the FCC and experiments